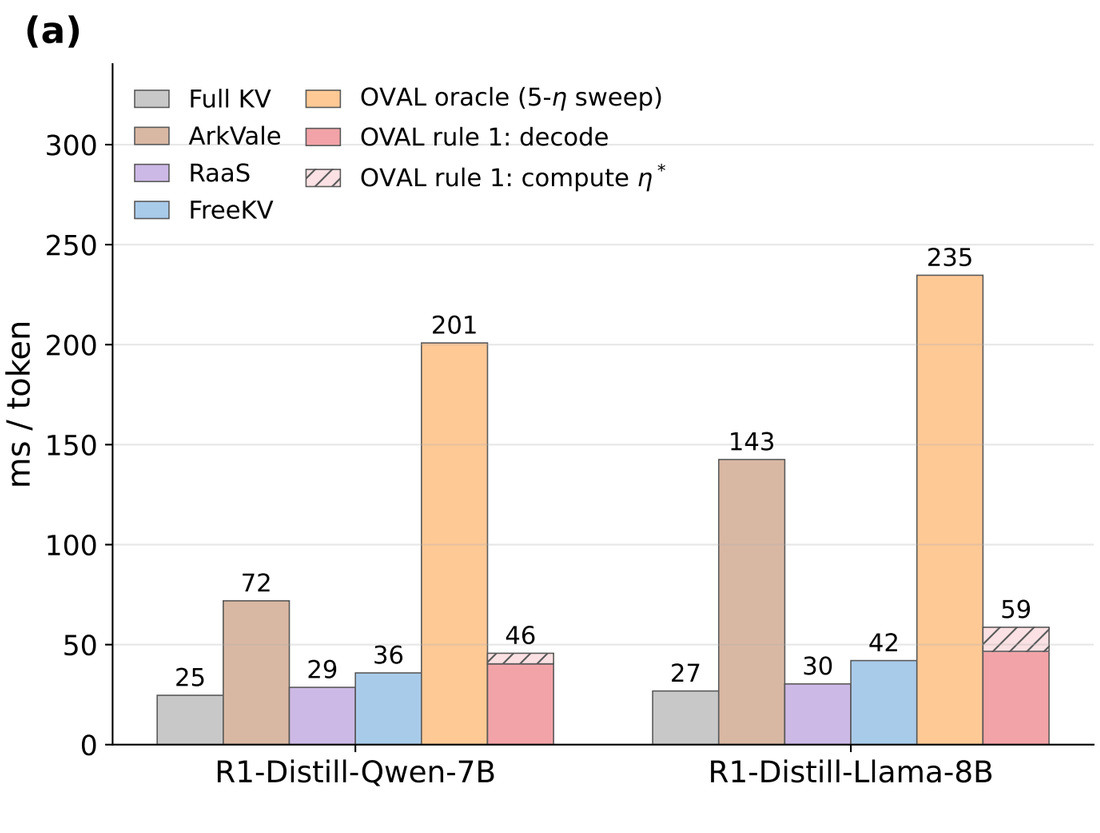
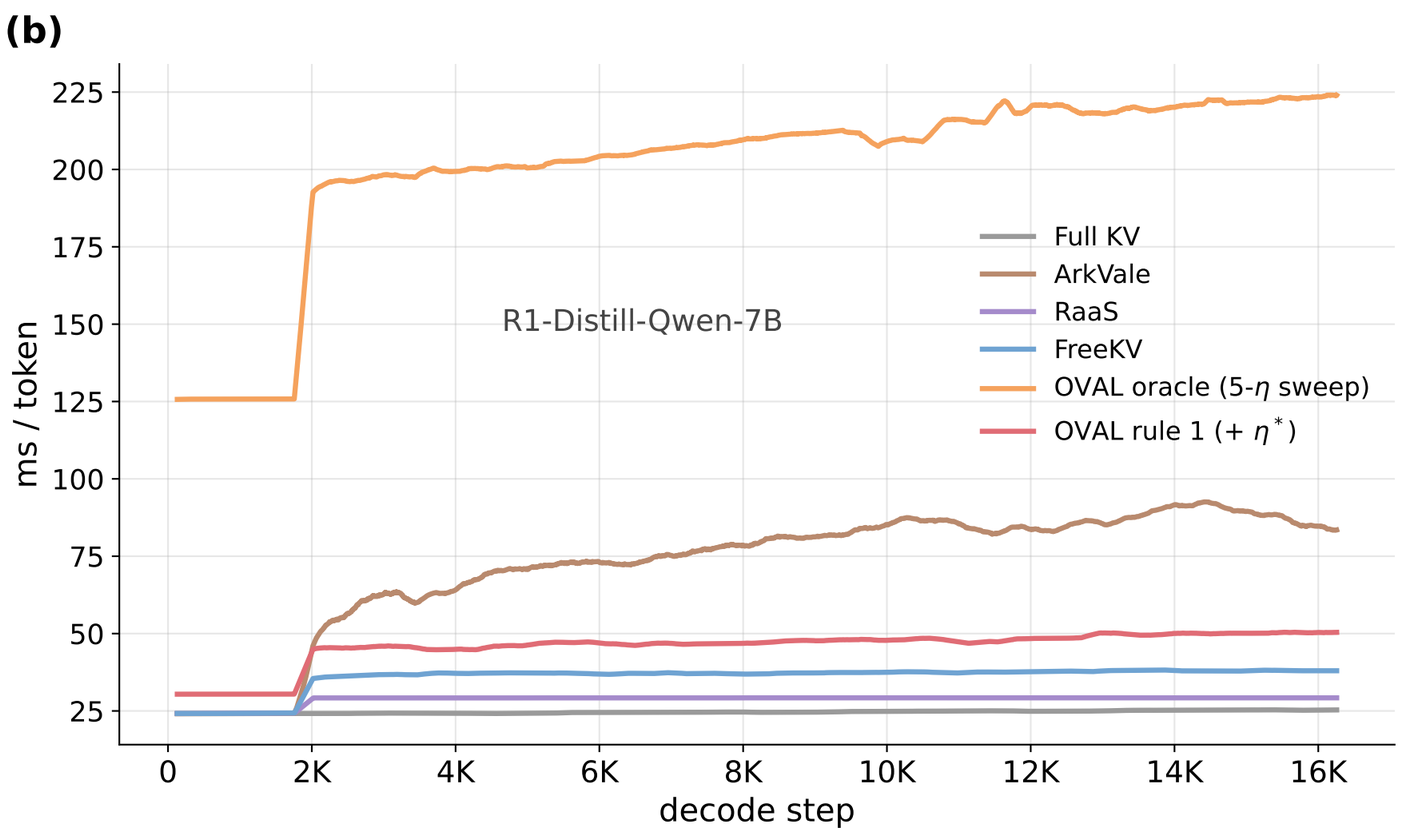
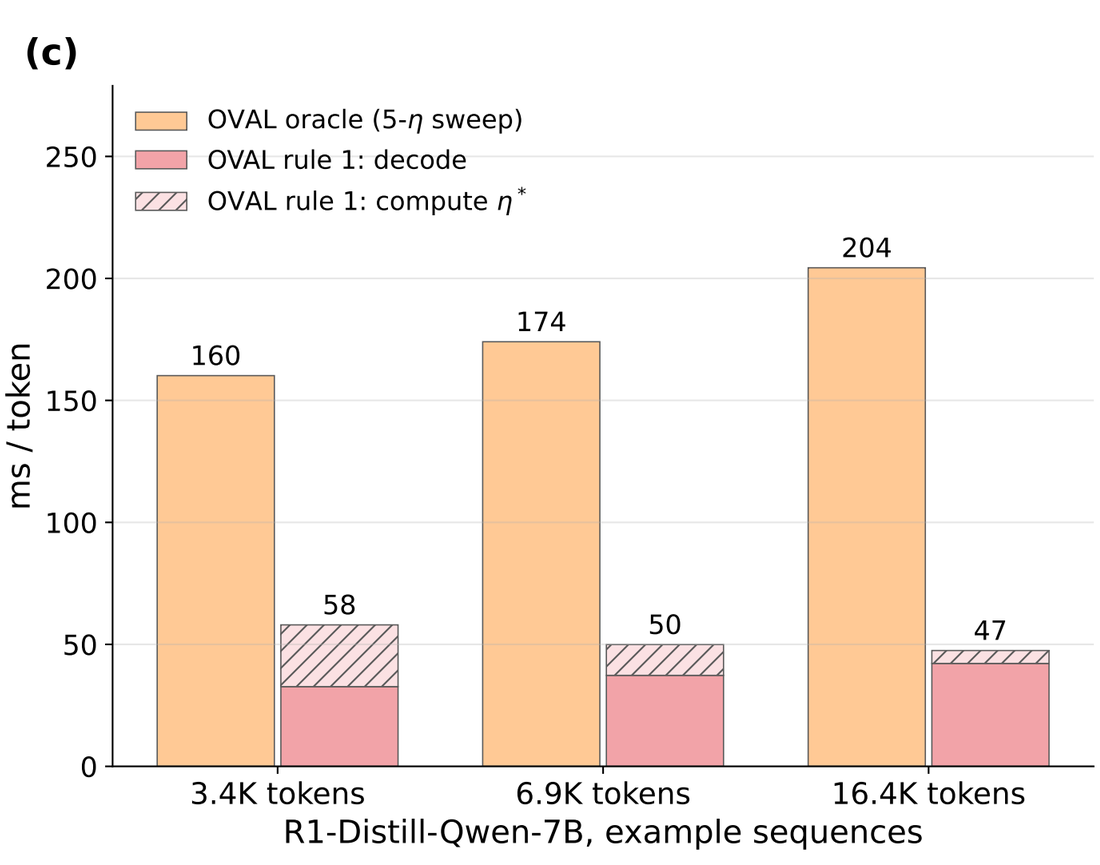
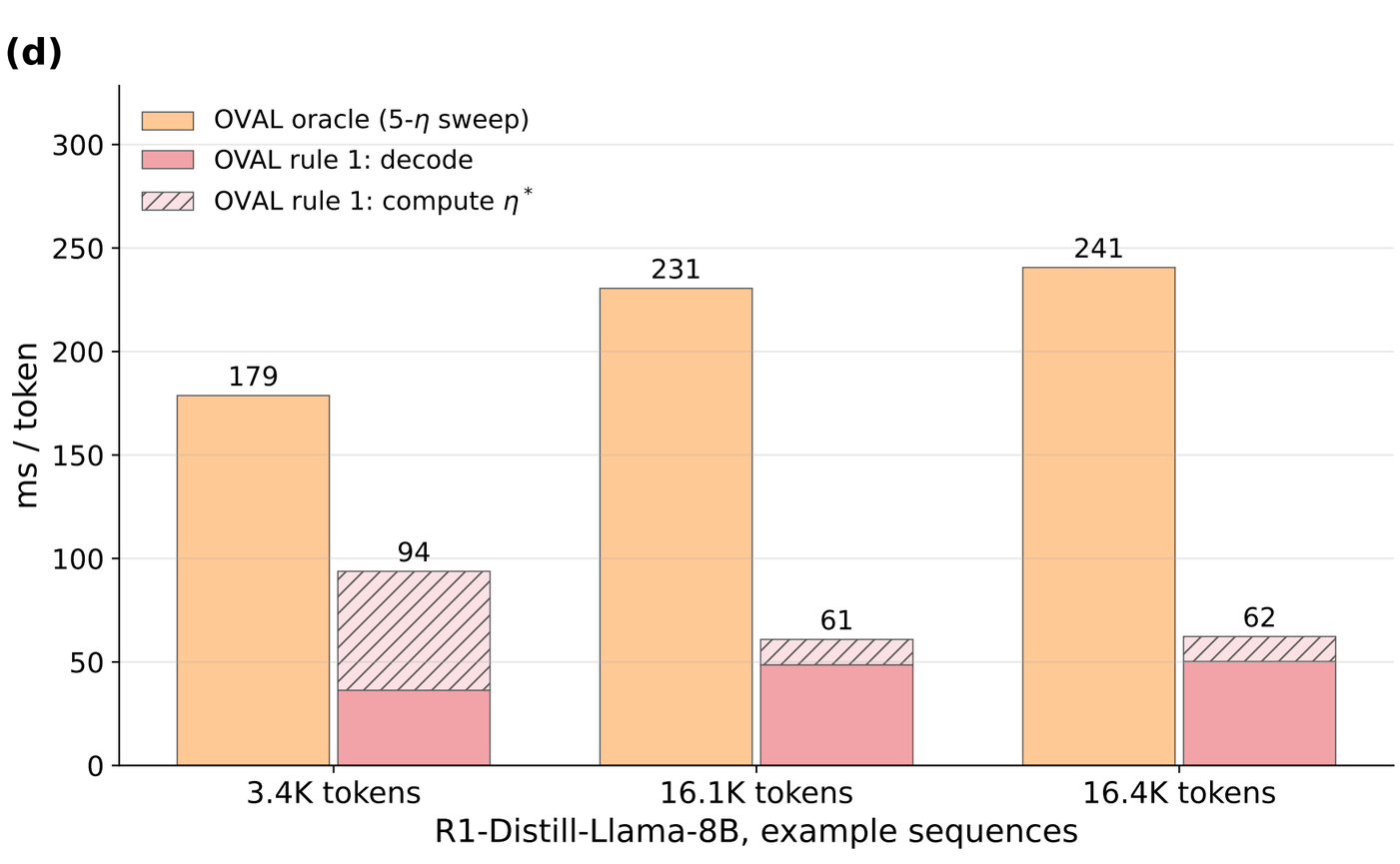

(a)
Full KV
ArkVale
RaaS
FreeKV
OVAL oracle (5-η sweep)
OVAL rule 1: decode
OVAL rule 1: compute η*
ms / token
0
50
100
150
200
250
300
25
72
29
36
201
46
27
143
30
42
235
59
R1-Distill-Qwen-7B
R1-Distill-Llama-8B


(b)
R1-Distill-Qwen-7B
Full KV
ArkVale
RaaS
FreeKV
OVAL oracle (5-η sweep)
OVAL rule 1 (+ η*)
ms / token
25
50
75
100
125
150
175
200
225
0
2K
4K
6K
8K
10K
12K
14K
16K
decode step


(c)
OVAL oracle (5-η sweep)
OVAL rule 1: decode
OVAL rule 1: compute η*
ms / token
0
50
100
150
200
250
160
58
174
50
204
47
3.4K tokens
6.9K tokens
16.4K tokens
R1-Distill-Qwen-7B, example sequences


(d)
OVAL oracle (5-η sweep)
OVAL rule 1: decode
OVAL rule 1: compute η*
ms / token
0
50
100
150
200
250
300
179
94
231
61
241
62
3.4K tokens
16.1K tokens
16.4K tokens
R1-Distill-Llama-8B, example sequences